

Evaluating the Impact of Monte Carlo Simulations on Cost-Benefit Analysis Outcomes in Business Cases

Introduction:

Cost-Benefit Analysis (CBA) is a crucial tool in assessing the financial viability of projects in business and economics. Traditional CBA methods often neglect the uncertainties inherent in real-world scenarios, potentially leading to suboptimal decision-making. This research aims to explore whether applying Monte Carlo simulations to existing CBA analyses would change the decisions made in past business cases, providing a more robust framework for evaluating project outcomes.

Background and Literature Review:

Monte Carlo simulations have been widely applied in various fields to model uncertainty and variability, offering a more comprehensive view of potential outcomes compared to deterministic methods. In the context of CBA, limited research exists on integrating Monte Carlo simulations. Key studies have demonstrated the advantages of probabilistic approaches in financial modeling, risk assessment, and decision analysis. However, the specific application of Monte Carlo simulations to CBA remains underexplored.

This research will address the following gaps:

- The lack of empirical studies applying Monte Carlo simulations to historical business cases in CBA.
- Insufficient understanding of how probabilistic approaches can influence decision-making in project appraisals.
- The need for a standardised methodology to incorporate Monte Carlo simulations into CBA practices.

Research Methodology:

Data Sources: The study will use previously published business cases, IAs and Evaluations as primary data sources. These cases will be selected from reputable databases and cover various industries and time periods to ensure a comprehensive analysis.

Empirical Strategy: The research will employ a comparative analysis between traditional CBA and Monte Carlo-enhanced CBA. The following steps will be involved:

1. **Baseline Analysis:** Conduct traditional CBA on selected business cases to establish baseline results.
2. **Monte Carlo Simulation:** Apply Monte Carlo simulations to the same cases, incorporating randomness and variability into the analysis.